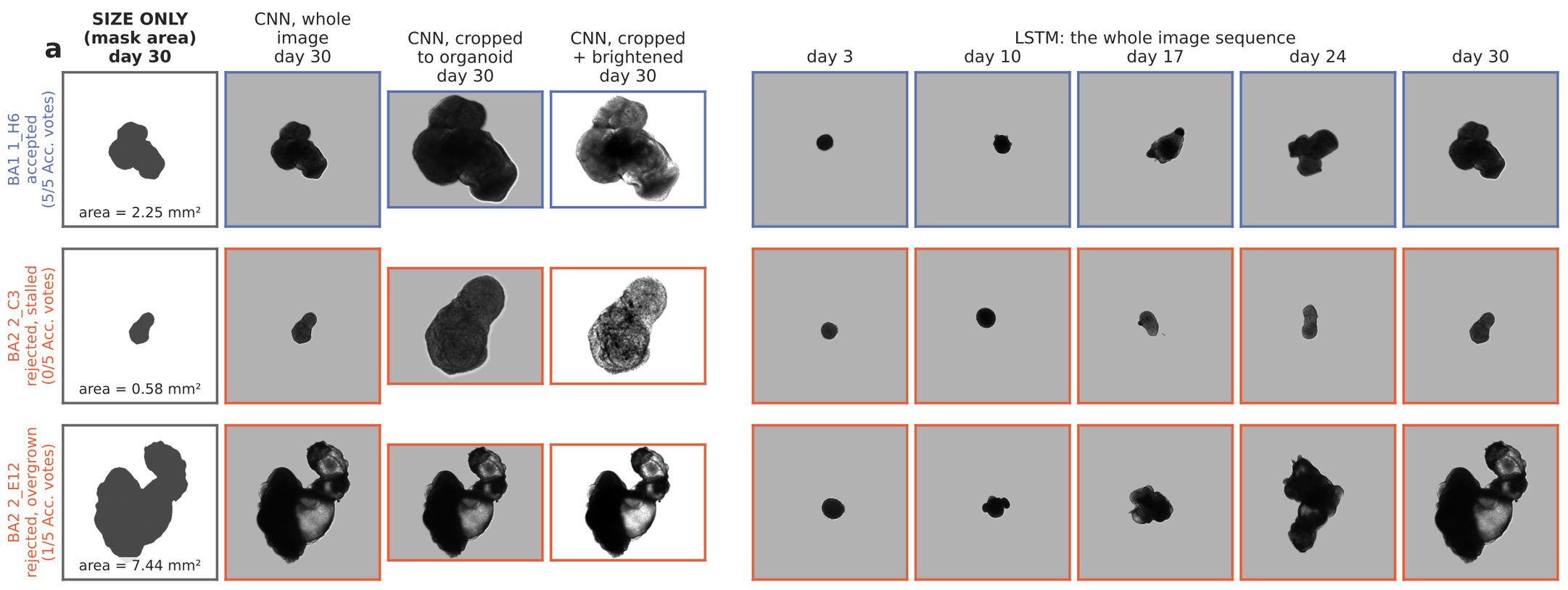
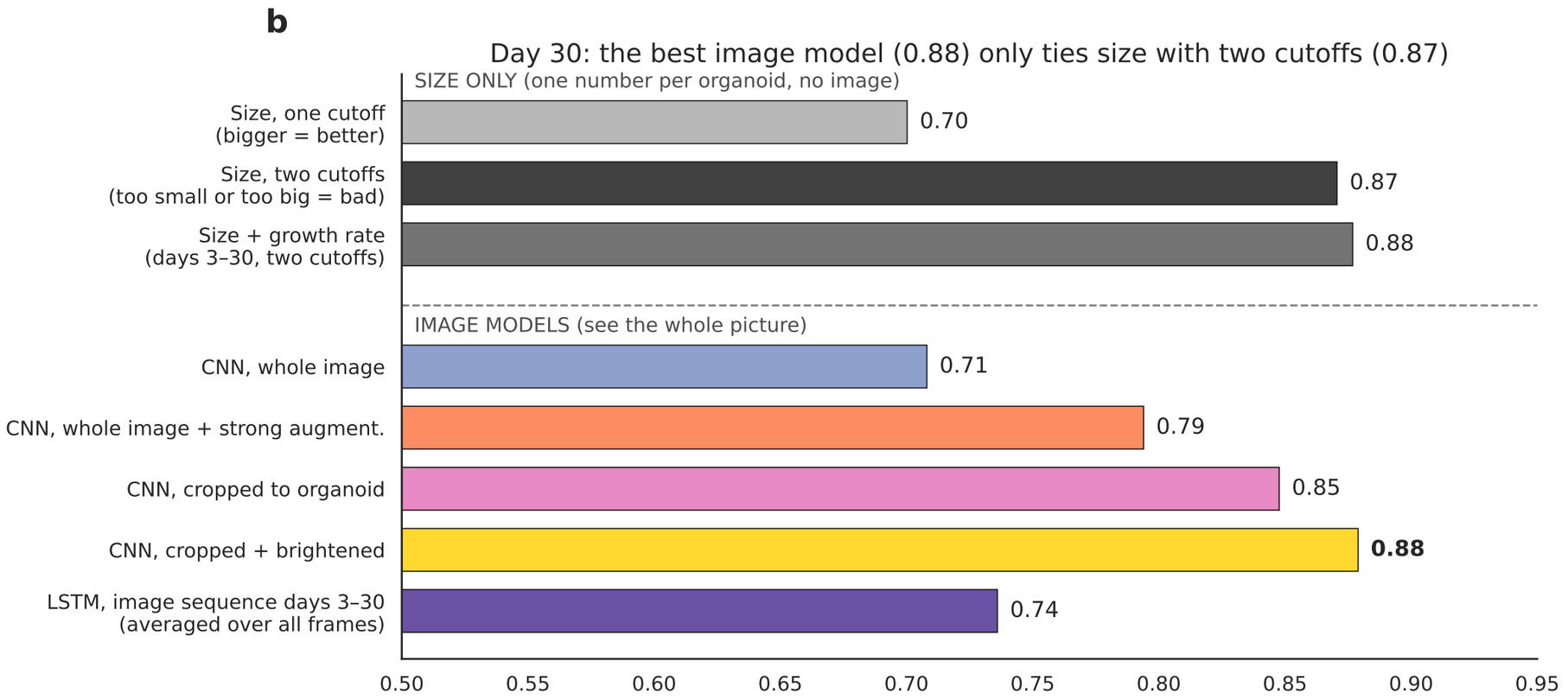
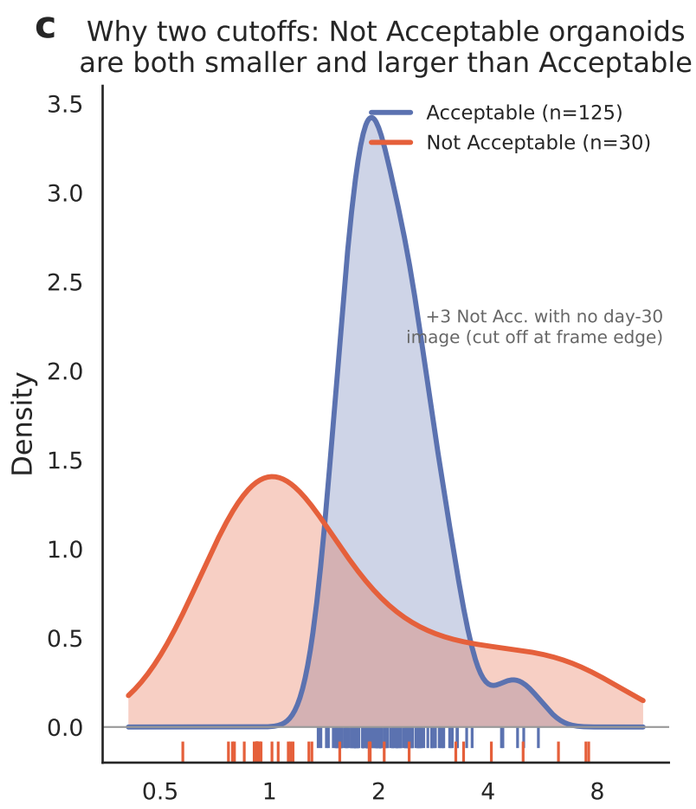
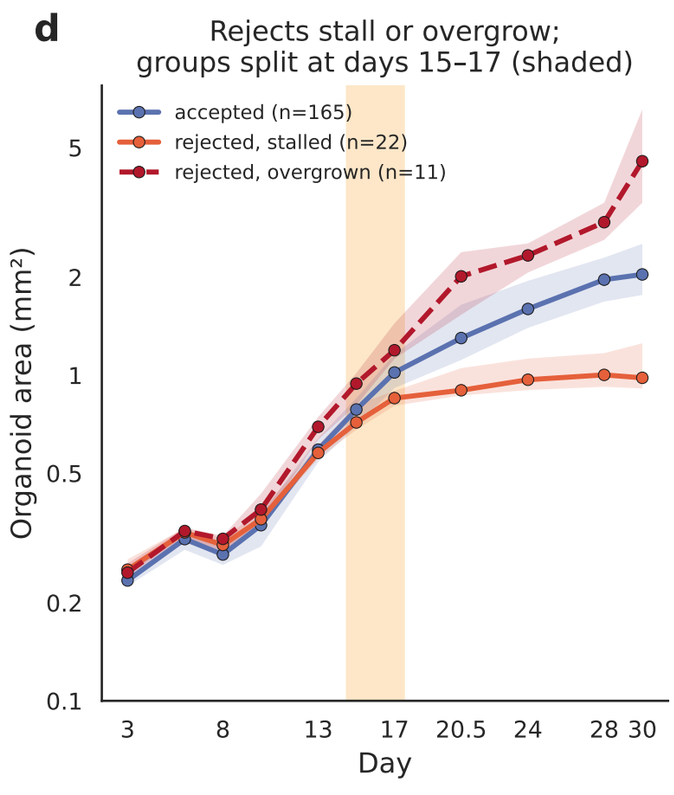
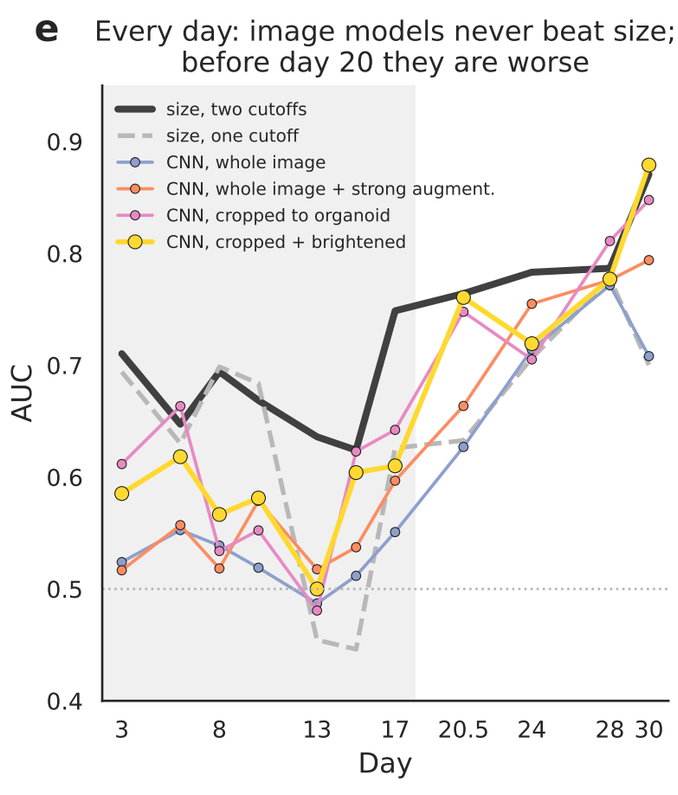
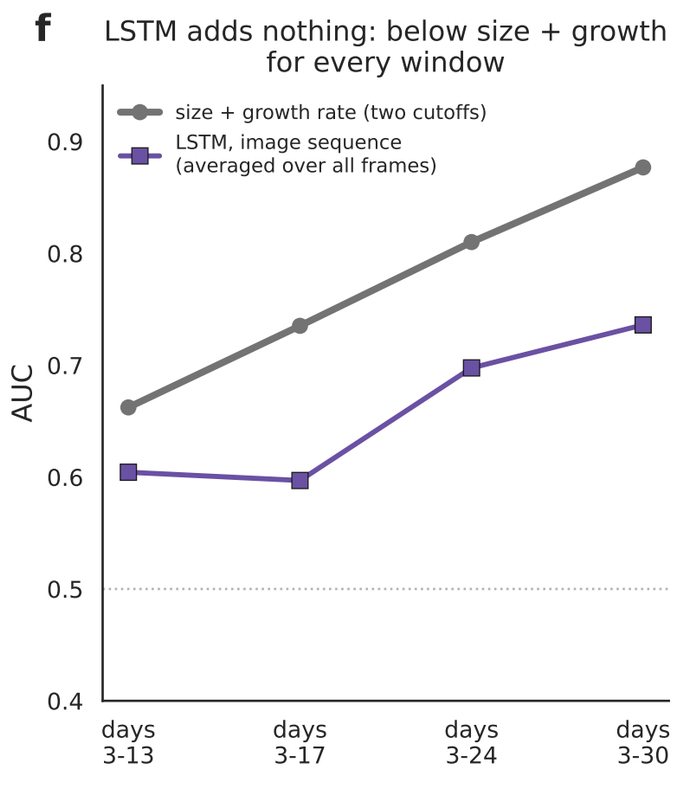

a
SIZE ONLY (mask area) day 30
CNN, whole image day 30
CNN, cropped to organoid day 30
CNN, cropped + brightened day 30
LSTM: the whole image sequence
day 3
day 10
day 17
day 24
day 30
BA1 1_H6 accepted (5/5 Acc. votes)
area = 2.25 mm²
BA2 2_C3 rejected, stalled (0/5 Acc. votes)
area = 0.58 mm²
BA2 2_E12 rejected, overgrown (1/5 Acc. votes)
area = 7.44 mm²


b
Day 30: the best image model (0.88) only ties size with two cutoffs (0.87)
SIZE ONLY (one number per organoid, no image)
Size, one cutoff (bigger = better)
0.70
Size, two cutoffs (too small or too big = bad)
0.87
Size + growth rate (days 3–30, two cutoffs)
0.88
IMAGE MODELS (see the whole picture)
CNN, whole image
0.71
CNN, whole image + strong augment.
0.79
CNN, cropped to organoid
0.85
CNN, cropped + brightened
0.88
LSTM, image sequence days 3–30 (averaged over all frames)
0.74
0.50 0.55 0.60 0.65 0.70 0.75 0.80 0.85 0.90 0.95
AUC at day 30 (0.5 = chance; mean of 4-fold × 10-repeat CV, same 155 organoids)


c
Why two cutoffs: Not Acceptable organoids are both smaller and larger than Acceptable
Acceptable (n=125)
Not Acceptable (n=30)
+3 Not Acc. with no day-30 image (cut off at frame edge)
Density
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.5 1 2 4 8
Day-30 organoid area (mm²)


d
Rejects stall or overgrow; groups split at days 15–17 (shaded)
accepted (n=165)
rejected, stalled (n=22)
rejected, overgrown (n=11)
Organoid area (mm²)
5
2
1
0.5
0.2
0.1
3 8 13 17 20.5 24 28 30
Day


e
Every day: image models never beat size; before day 20 they are worse
size, two cutoffs
size, one cutoff
CNN, whole image
CNN, whole image + strong augment.
CNN, cropped to organoid
CNN, cropped + brightened
AUC
0.9
0.8
0.7
0.6
0.5
0.4
3 8 13 17 20.5 24 28 30
Day


f
LSTM adds nothing: below size + growth for every window
size + growth rate (two cutoffs)
LSTM, image sequence (averaged over all frames)
AUC
0.9
0.8
0.7
0.6
0.5
0.4
days 3-13
days 3-17
days 3-24
days 3-30